* Verniers Scale:-

→ Verniers is a device for measuring accurately the fractional part of the smallest division on a graduated scale.

→ The actual reading is given by

[Reading on main scale before index mark + N × L.C]

N = No° of graduations on vernier scale

LC = Least Count

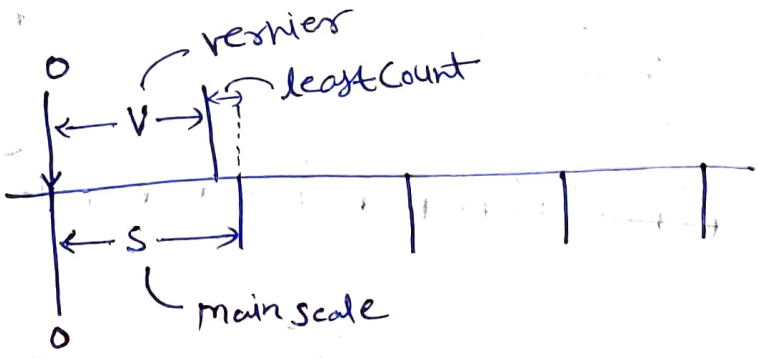
= 1 division on main scale − 1 division on vernier scale.

Types of verniers

1) Direct vernier:-

→ This vernier is calibrated in the same direction of main scale.

→ This vernier has division slightly smaller than division of main scale.



[V < S]

$$\boxed{\text{Least Count} = S - V}$$

S = length of one division of main scale
[least count of main scale]

V = length of on division of vernier scale
[Not the least count].

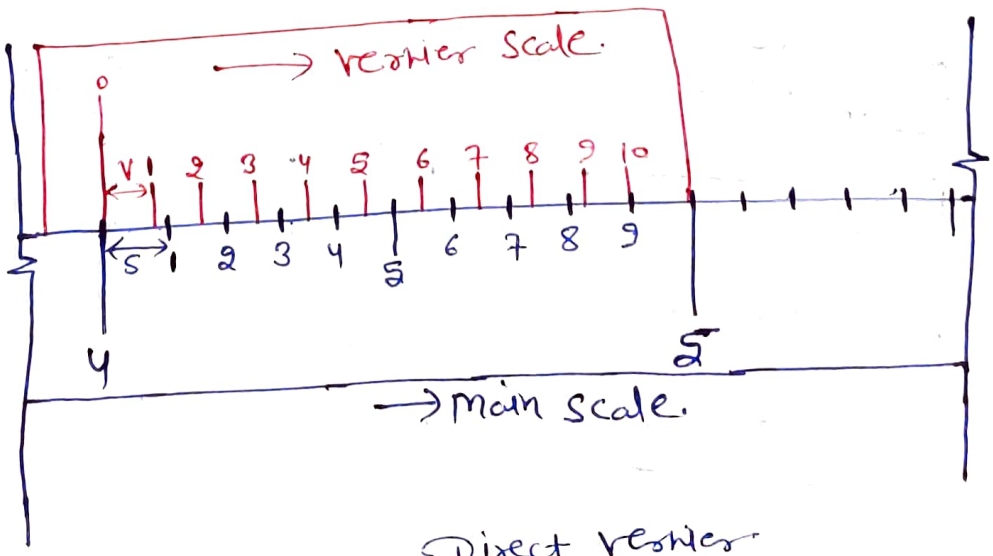
(n-1) division of main scale is distributed into n division of vernier scale.

$$nV = (n-1)S$$

$$V = \left(\frac{n-1}{n}\right).S$$

$$\text{Least Count} = S - V$$
$$= S - \frac{n-1}{n}.S$$
$$= S\left[1 - \frac{n-1}{n}\right]$$
$$= S\left[\frac{\cancel{n}-\cancel{n}+1}{n}\right]$$
$$= \frac{S}{n}$$

$$\boxed{\text{Least Count} = S/n}$$



Direct vernier

ii). Retrograde vernier:-

→ This vernier is calibrated in the opposite direction of main scale.

→ This vernier has division slightly longer than division of main scale [V > S].

→ (n+1) division of main scale is divided into n-division of vernier scale.

$$n \cdot V = (n+1) \cdot S$$

$$V = \frac{n+1}{n} \cdot S$$

Now least Count $= V - S$

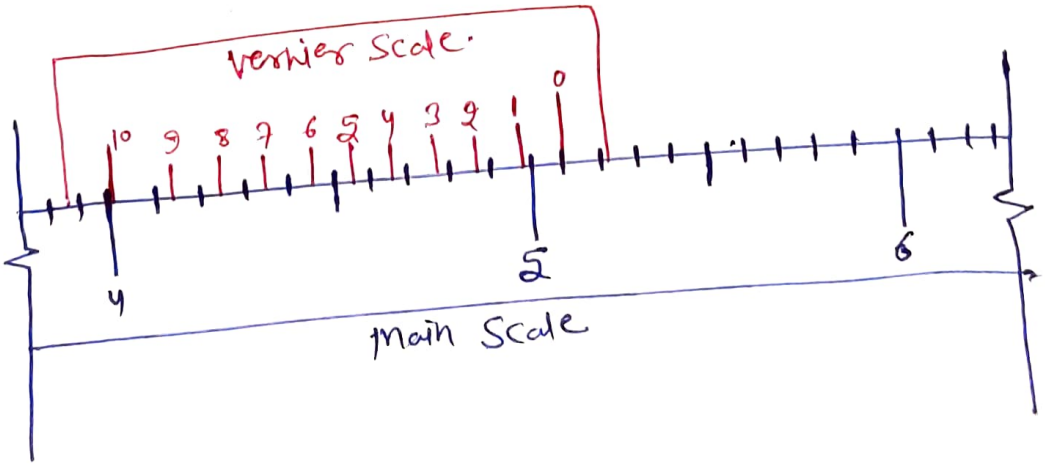
$$= \frac{n+1}{n} S - S$$

$$= S\left[\frac{n+1}{n} - 1\right]$$

$$= S\left[\frac{\cancel{n}+1-\cancel{n}}{n}\right]$$

$$= S/n$$

$$\boxed{L.C = \frac{S}{n}}$$



iii) Extended vernier:-

→ When (2n-1) division of main scale is divided into n division of vernier scale then it is called extended vernier.

iv). Double vernier:-

→ When the vernier is calibrated in the both direction of index line it is called double vernier.

→ Double vernier consist of two direct verniers extending in both directions with the index mark in the centre.

→ It is combination of both direct and retrograde vernier.

→ To use this vernier scale main scale should also be calibrated in both direction.

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